

Soln 3:  $A \rightarrow \text{spy}, B \rightarrow \text{Knight}, C \rightarrow \text{Knave}$

Soln 4:  $A \rightarrow \text{spy}, B \rightarrow \text{Knave}, C \rightarrow \text{Knight}$

Soln 5:  $A \rightarrow \text{Knave}, B \rightarrow \text{spy}, C \rightarrow \text{Knight}$

Soln 6:  $A \rightarrow \text{Knave}, B \rightarrow \text{Knight}, C \rightarrow \text{spy}$

34) A says "I am not the spy," B says "I am not the spy," and C says "A is the spy."

**Soln:** A Knave can never say "I am not the spy" since it is True and a Knave must always lie.  $\therefore$  A and B cannot be Knaves

By problem constraint, C is the Knave. C says "A is the spy" which is False.  $\therefore$  A must be the Knight. A says "I am not the spy" which is True and consistent with the fact that a Knight always speaks the truth. So, B must be the spy. B says "I am not the spy," which is False and consistent with the fact that a spy can lie sometimes.

Unique solution:  $A \rightarrow \text{Knight}, B \rightarrow \text{spy}, C \rightarrow \text{Knave}$

35) A says "I am not the spy," B says "I am not the spy", and C says "I am not the spy."

**Soln:** By soln. of problem 34, we know that a Knave can never say "I am not the spy." From problem constraints, since exactly one of A, B or C must be a Knave, so No solution exists in this situation.

Exercises 36-42 are puzzles that can be solved by translating statements in logical expressions and reasoning from these expressions using truth tables.

36) The police have three suspects for the murder of Mr. Cooper: Mr. Smith, Mr. Jones and Mr. Williams. Smith, Jones and Williams each declare that they did not kill Cooper. Smith also states that Cooper was a friend of Jones and that Williams disliked him. Jones also states that he did not know